

Kristine Stanley

Staff / Senior Python AI/ML & LLM Engineer

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PROFESSIONAL SUMMARY

Staff/Senior Python AI/ML Engineer with 11+ years building production intelligence platforms across application security, connected mobility, device protection, and SaaS using **Python 3.4–3.13**, **PyTorch**, **TensorFlow**, **scikit-learn**, **FastAPI**, **vector databases**, **AWS**, and **Kubernetes**, delivering LLM applications, NLP pipelines, predictive models, MLOps automation, scalable APIs, and measurable gains in accuracy, reliability, and engineering productivity.

SKILLS

- **Programming:** Python, SQL, Bash, TypeScript, JavaScript, Java, object-oriented programming, functional programming, asynchronous programming, multiprocessing
- **Machine Learning:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, classification, regression, clustering, forecasting, anomaly detection, ranking, recommendation systems
- **Generative AI and LLMs:** OpenAI APIs, Azure OpenAI, AWS Bedrock, Anthropic Claude, Hugging Face Transformers, LangChain, LlamaIndex, prompt engineering, tool calling, structured outputs
- **LLM Engineering:** Retrieval-augmented generation, hybrid search, reranking, embeddings, chunking, semantic caching, function calling, agentic workflows, LoRA, PEFT, model evaluation, guardrails
- **NLP:** Text classification, named-entity recognition, intent detection, semantic similarity, summarization, keyword extraction, topic modeling, tokenization, sentiment analysis
- **Data Engineering:** Pandas, NumPy, PySpark, Apache Spark, Apache Kafka, Airflow, AWS Glue, ETL/ELT, feature engineering, batch processing, streaming pipelines
- **Model Serving:** FastAPI, Flask, Pydantic, Celery, BentoML, MLflow, TorchServe, asynchronous APIs, batch inference, real-time inference, model versioning
- **Vector and Search Platforms:** pgvector, Pinecone, FAISS, OpenSearch, Elasticsearch, Chroma, vector indexing, metadata filtering, hybrid retrieval
- **MLOps:** MLflow, Kubeflow, DVC, Docker, Kubernetes, Helm, Terraform, GitHub Actions, Jenkins, model registry, experiment tracking, automated retraining
- **Cloud Platforms:** AWS SageMaker, Bedrock, Lambda, EKS, ECS, S3, RDS, DynamoDB, SQS, CloudWatch, Azure Machine Learning, Google Vertex AI
- **Databases:** PostgreSQL, MySQL, SQL Server, MongoDB, DynamoDB, Redis, data modeling, indexing, query optimization, feature storage
- **Testing and Quality:** Pytest, unittest, Hypothesis, integration testing, contract testing, data validation, Great Expectations, model evaluation, prompt regression testing
- **Monitoring:** OpenTelemetry, Prometheus, Grafana, Evidently, Arize, Datadog, CloudWatch, model drift, data quality, latency, cost and token monitoring
- **AI Security and Governance:** PII redaction, prompt-injection defenses, access control, audit logging, responsible AI, explainability, bias testing, human-in-the-loop review
- **Engineering Leadership:** AI architecture, technical roadmaps, design reviews, ADRs, code reviews, mentoring, incident response, Agile delivery, stakeholder communication

EXPERIENCE

Veracode Systems Inc — Senior Software Engineer

Remote/ Full time

April 2021 – May 2026

Somerville, MA

- Architected production AI services with **Python 3.8–3.13**, **FastAPI**, **PyTorch**, and **PostgreSQL** for vulnerability classification, scan-result enrichment, and risk-based application security workflows.
- Built supervised learning models that combined static-analysis findings, exploitability signals, asset criticality, and historical remediation behavior to prioritize vulnerabilities for enterprise security teams.

- Introduced a secure **retrieval-augmented generation platform** using transformer embeddings, vector search, prompt templates, reranking, and source citations to explain findings and recommend remediation steps.
- Reduced average analyst triage time by **42%** through calibrated risk scoring, semantic deduplication, automated finding summaries, and human-in-the-loop feedback captured from security reviewers.
- Resolved a production defect in which mismatched tokenizer and embedding-model releases returned incomplete retrieval results by pinning artifacts, validating dimensions, and adding canary evaluations.
- Designed repeatable **MLOps pipelines** with MLflow, Airflow, Docker, Kubernetes, and AWS to manage training data, experiments, model registration, approval gates, deployment, and rollback.
- Modernized legacy Python workers and notebook-driven experiments into typed, testable packages with dependency locking, asynchronous processing, standardized configuration, and reusable feature components.
- Developed low-latency REST and asynchronous inference APIs with **FastAPI, Redis, Kafka, and Celery**, supporting batch scoring, streaming enrichment, rate limits, and graceful degradation.
- Implemented model and platform observability with **OpenTelemetry, Prometheus, Grafana, Evidently, and CloudWatch** to monitor latency, drift, feature quality, token usage, and prediction distributions.
- Led AI architecture reviews, responsible-use assessments, incident retrospectives, code reviews, and mentoring while aligning security, product, data, and platform teams around measurable outcomes.

Cubic Telecom USA — Software Engineer

Remote/ Full time

January 2019 – April 2021

San Diego, CA

- Developed data and machine learning services with **Python 3.6–3.9, scikit-learn, TensorFlow, Spark, and Kafka** for connected-vehicle telemetry, network quality, and subscription operations.
- Created anomaly-detection pipelines that identified abnormal roaming behavior, modem resets, usage spikes, and carrier connectivity degradation across high-volume IoT event streams.
- Reduced false-positive operational alerts by **34%** by combining robust statistical thresholds, device-segment baselines, temporal features, and feedback from network operations specialists.
- Corrected a production scoring issue caused by late and out-of-order telemetry by introducing event-time processing, bounded windows, deduplication keys, and replay-safe feature generation.
- Built demand and usage forecasting models that helped capacity teams anticipate regional traffic patterns, carrier consumption, and subscription growth for connected mobility programs.
- Established reusable feature-engineering components for device history, geography, network state, and subscription behavior, reducing inconsistent logic between training and inference pipelines.
- Orchestrated batch and streaming workloads with **Airflow, Spark, Kafka, PostgreSQL, and S3** while adding data-quality checks, schema validation, lineage metadata, and failure recovery.
- Containerized inference services with Docker and Kubernetes on AWS, adding autoscaling, health checks, model versioning, blue-green deployment controls, and centralized monitoring.
- Partnered with telecom engineers, product managers, analysts, and SRE teams to translate operational problems into measurable machine learning use cases and production-ready services.

Asurion — Software Developer

Remote/ Full time

January 2017 – December 2018

Nashville, TN

- Built predictive services with **Python 3.5–3.7, scikit-learn, XGBoost, Flask, and PostgreSQL** for device-protection claims, repair routing, fraud screening, and support operations.
- Developed classification models using claim history, device attributes, fulfillment patterns, and customer interactions to prioritize cases for automated handling or specialist review.
- Reduced unnecessary manual claim reviews by **27%** through threshold tuning, probability calibration, targeted feature engineering, and clear routing rules for low-confidence predictions.

- Implemented NLP pipelines for support-ticket classification, intent detection, and issue tagging using TF-IDF, word embeddings, and supervised models trained on historical service conversations.
- Detected and removed a data-leakage defect involving post-resolution fields that inflated offline accuracy, then rebuilt time-aware datasets and added feature-availability validation.
- Exposed model predictions through versioned **Flask APIs and Celery workers** with request validation, timeout controls, audit logging, retry policies, and fallback business rules.
- Automated training and scoring workflows with Airflow, SQL, and Jenkins while monitoring missing data, class imbalance, model performance, and upstream schema changes.
- Worked with claims specialists, data analysts, software engineers, and compliance partners to review model behavior, explain decisions, investigate production errors, and release improvements.

Rustici Software — Junior Software Developer

Hybrid/ Full time

June 2015 – December 2016

Nashville, TN

- Developed data-processing utilities with **Python 2.7 and Python 3.4–3.5, Flask, Celery, PostgreSQL, and MySQL** for e-learning content ingestion, tracking, and reporting workflows.
- Created parsers and validation pipelines for SCORM packages, xAPI statements, metadata, and learner activity records while preserving compatibility with diverse customer content.
- Built classical NLP features for course tagging, keyword extraction, duplicate-content detection, and search relevance using tokenization, TF-IDF, and similarity scoring.
- Resolved recurring import failures caused by malformed XML, inconsistent character encodings, oversized archives, and invalid timestamps by adding defensive parsing and actionable diagnostics.
- Modernized ad hoc scripts and legacy processing jobs into reusable Python modules and queue-based workers with structured logging, configuration management, and predictable error handling.
- Added unit tests, fixture-based parser tests, integration coverage, and CI checks that reduced regression risk across content-processing and reporting components.
- Collaborated with senior developers and customer-support specialists to reproduce production issues, document root causes, refine requirements, and deliver maintainable platform enhancements.

EDUCATION

Tennessee State University — Bachelor's Degree in Computer Science

Sep 2011 – May 2015

CERTIFICATION

- **AWS Certified Machine Learning – Specialty**
- **Google Cloud Professional Machine Learning Engineer**
- **Databricks Certified Machine Learning Professional**